

SECURE CODING REVIEW

CODEALPHA CYBER SECURITY INTERNSHIP

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Introduction :

Secure coding is the practice of developing software in a way that protects applications from security vulnerabilities and cyber attacks. The purpose of this review is to identify common coding mistakes and recommend secure coding practices.

Vulnerable Code Example

```
username = input("Enter username: ")  
query = "SELECT * FROM users WHERE username = '" + username + "'"
```

Security Vulnerability Identified

SQL Injection

The above code directly inserts user input into an SQL query.

Risks:

- Unauthorized access
- Data theft
- Data modification
- Database compromise

Secure Code Example :

```
username = input("Enter username: ")

cursor.execute(
    "SELECT * FROM users WHERE username = %s",
    (username,)
)
```

Recommendations :

- Validate all user inputs
- Use parameterized queries
- Store passwords securely
- Encrypt sensitive data
- Use proper error handling

Conclusion :

Secure coding practices help developers create safer applications by reducing vulnerabilities and protecting user data from cyber threats.